ME 144L LE 6: Strain Gauges and Force Measurement

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1 Load Cell Calibration

1.1 Calibration Procedure and Data

A strain-gauge mini load cell was calibrated by suspending known masses (0, 10, 20, 50, and 100 g) and recording the corresponding integer output from the Arduino UNO. A linear regression was then performed to map raw counts to mass in grams. The calibration data are shown in Table 1.

Table 1: Raw Load Cell Readings vs. Calibration Masses

Mass [g]	**Load Cell Reading [int counts]**
0	416,888
10	446,004
20	475,515
50	563,700
100	710,520

These values can be plotted into the `CalibrateLoadCell.py` program, where the mass readings are a function of raw integer counts read from the Arduino serial monitor. A linear regression (Mass $= m \cdot \mathrm{LC} + b$) was used for the following result, where x represents the raw integer counts:

$$\text{Mass [g]} = 3.40 \times 10^{-4} \cdot x - 141.878 \tag{1}$$

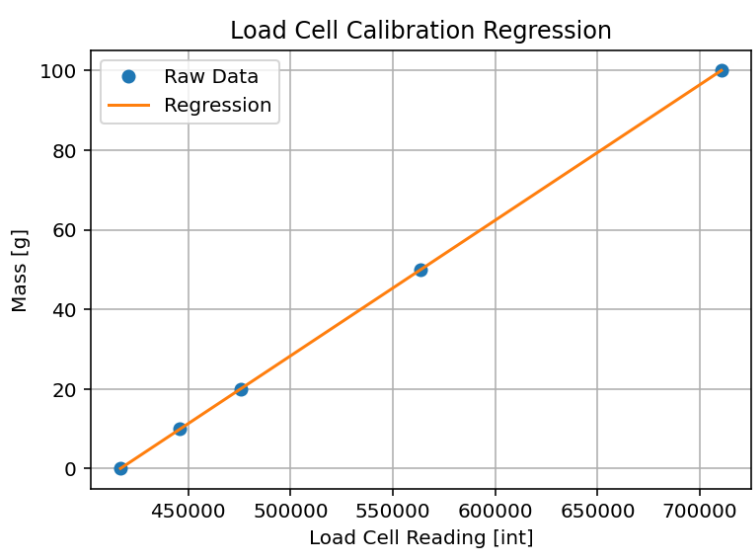

Figure 1: Load Cell Calibration Regression: Mass vs. Raw Count with Linear Fit

With an R^2 value of 0.9999, this indicates a strong linear fit. To express this relation in Newtons, Equation 1 is converted using $F = (m_{\text{grams}}) \times (1\,\text{kg}/1000\,\text{g}) \times g$, where g is the gravitational constant of 9.81 $\frac{\text{m}}{\text{s}^2}$.

$$F = \left(3.40 \times 10^{-4} \cdot x - 141.878\right) \times 10^{-3} \times 9.81 \tag{2}$$

The sensitivity of the load cell system is defined as the force increment, per raw count increment. From the previously derived Force equation, the sensitivity equation will be simplified in the form of Newtons per count:

$$S = 3.34 \times 10^{-6}\ \frac{\text{N}}{\text{count}} \tag{3}$$

This sensitivity means a change of approximately 300,000 counts per 1 N change.

1.2 Calibration Verification

The calibration was verified by placing the 50 g, 100 g, and 200 g reference weights on the load cell and confirming the readout after applying the calibrated load cell mass equation. The results from the table below agree with the expected gravitational force of $m \cdot g = 0.09994 \times 9.81 = 0.980\,\text{N}$, a percent error of less than 0.5%. This confirms the calibration equation is valid over the measured range and that the sensor linearity is excellent.

Table 2: Calibration Verification Data

Actual Mass (g)	**Int Reading**	**Measured Mass (g)**	**Percent Error**
50	563,628	49.755	0.490%
100	710,821	99.800	0.200%
200	1,003,404	199.278	0.361%

2 Bungee Cord Stiffness Characterization

2.1 Force vs. Displacement Measurement

The bungee cord was stretched in 10 mm increments from 0 to 120 mm, and the corresponding load cell integer output was recorded at each displacement. Equation 2 for solving calibrated force was then applied to convert the counts to Newtons. The resulting force-displacement data are tabulated in Table 2 and plotted in Figure 3.

Table 3: Calibrated Bungee Cord Force vs. Stretch Data

Stretch [mm]	**LC Reading [int]**	**Force [N]**
0	507,386	0.30
10	1,357,956	3.14
20	1,600,443	3.95
30	1,748,936	4.44
40	1,887,860	4.90
50	2,002,329	5.29
60	2,116,206	5.67
70	2,183,967	5.89
80	2,314,433	6.33
90	2,366,345	6.50
100	2,484,102	6.89
110	2,566,164	7.17
120	2,647,827	7.44

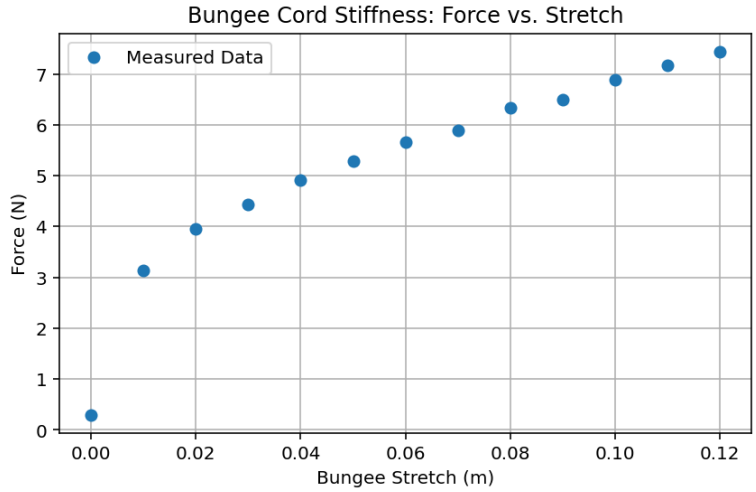

Figure 2: Bungee Cord Force vs. Stretch Plot

2.2 Discussion of Bungee Stiffness Behavior

The force-displacement curve in Figure 2 shows nonlinear behavior. At small displacements (0–10 mm), the force changes drastically from approximately 0.30 N to 3.14 N, indicating a very high initial tangent stiffness. This is likely a result of how the experimentation was performed for finding the zero strain force applied, which was done with a 10g weight, leaving slack in the bungee cord. Beyond 10 mm, the curve becomes more gradual and approximately linear. For the purposes of determining stiffness (slope), it is best to omit the first data point. Note that including the first data point results in a stiffness of 46.37 N/m. Using python `model.coef_` and `model.score()` tools on the trimmed dataset yields the following:

$$k_{\text{bungee}} = 36.65 \text{ N/m}, \quad R^2 = 0.977 \tag{4}$$

$$F = 36.653 \cdot x + 3.25 \tag{5}$$

Note that the process for determining the zero strain applied force could be improved, however the data cleaning process allows for more stiffness values that are much more representative of system dynamics.

3 Dynamic Force Measurement of Small Oscillations

3.1 Measured Force vs. Time

A nominal 100 g mass was attached to the bungee cord and stretched roughly 1/2 inches for a small oscillation test. The load cell recorded force while the mass was allowed to oscillate freely. Figure 3 shows the measured force as a function of time. The mass is released at $t \approx 0$ s, initiating oscillations that decay over a set 8 seconds. The equilibrium applied force force converges to approximately 0.983 N, consistent with $m \cdot g = 0.980$ N. Below is the recorded calibrated masses of the bungee and weight:

Table 4: Calibrated Masses of Bungee Cord and 100g Load

Object	**Symbol**	**Mass [g]**
100g Mass	m_{mass}	99.94 g
Bungee	m_{bungee}	10.81 g

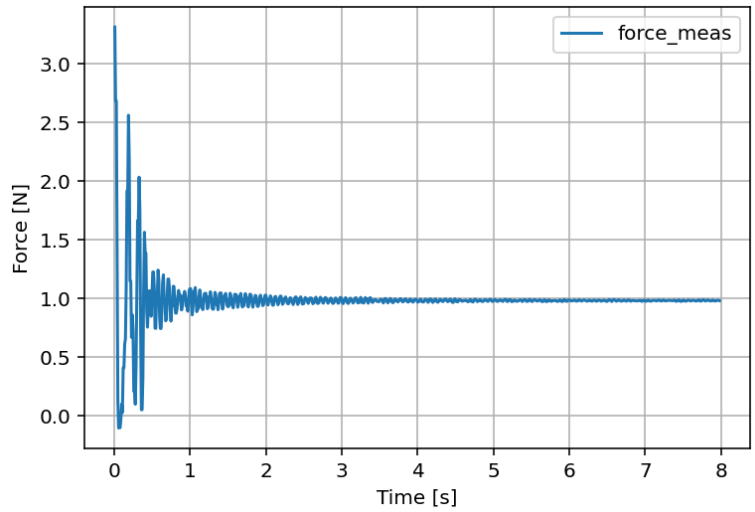

Figure 3: Measured Force vs. Time During Small Oscillations of the 100 g Mass on the Bungee Cord

The measured force signal in Figure 3 represents underdamped oscillatory behavior following the initial disturbance. The oscillations decay to within 5% of equilibrium by approximately $t \approx 2\,\mathrm{s}$, reflecting the lightly damped system dynamics.

The early cycles are visibly non-sinusoidal, which is likely due to the asymmetric restoring force of the bungee cord. On the upstroke, when the mass displaces above the equilibrium position, the cord gets slack and provides no restoring force, relying on just gravity. Tension is only recovered on the downstroke once the cord re-engages, producing an impulse rather than a smooth sinusoidal return. After about 1 second, these effects seem to dissipate, and amplitude decays and the mass remains within bungee tension, resulting in a more regular decaying sinusoidal motion.

4 Comparison of Measured Force and Acceleration-Derived Force

4.1 Measurement Overview

For a second trial, an accelerometer was attached to the bottom of the 100g applied mass. In this experiment, both the load cell force and the IMU accelerometer Z-axis were recorded simultaneously. Multiple trials were done before the lab group converged on this solution, as we determined it to have the least lateral motion or rotation that would impact accelerometer and load cell readings. The acceleration-derived force is computed as:

$$F_{\mathrm{acc}}(t) = -m \cdot a_z(t) \tag{6}$$

where $m = 99.94\,\mathrm{g}$ and a_z is the Z-axis acceleration in $\mathrm{m/s^2}$. The accelerometer raw counts are then converted using the spec for 2g testing:

$$a_z\ [\mathrm{m/s^2}] = z_a \cdot (2)(0.061 \times 10^{-3})(9.81) \tag{7}$$

4.2 Results and Discussion

Figure 4 shows both F_{meas} from the load cell and $F_{\mathrm{acc}} = -ma_z$ on the same axes, along with the raw acceleration in the lower panel.

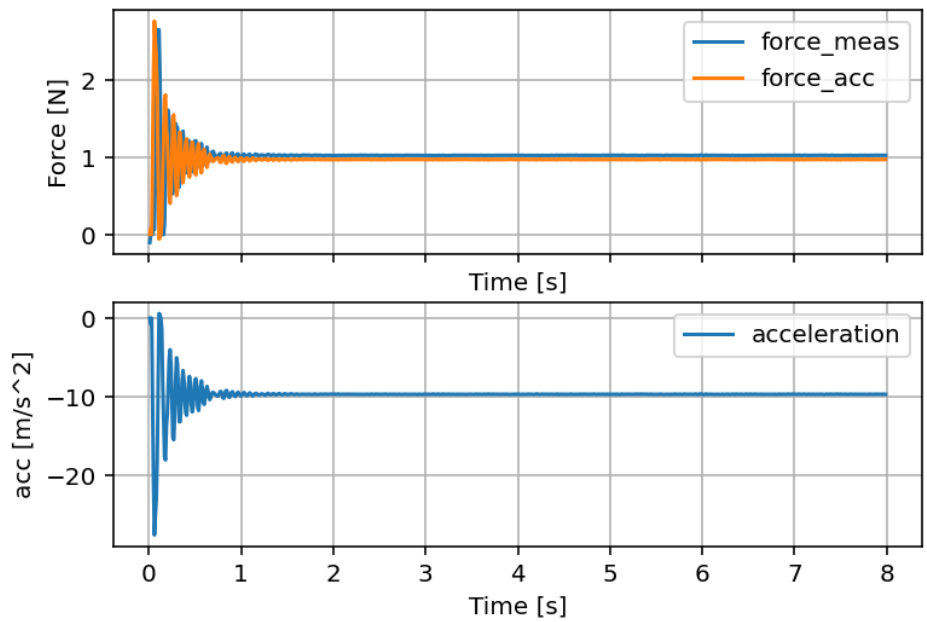

Figure 4: Top: Measured Load Cell Force and Acceleration-Derived Force vs. Time. Bottom: Z-Axis Acceleration vs. Time

The two signals align closely in Figure 4, which is expected from Newton's 2nd Law in equation 6. Because the Accelerometer was oriented with the Z-axis normal facing down, in the steady state, $a_z \approx -g = -9.81\,\mathrm{m/s^2}$, and $F_{\mathrm{acc}} = -m(-g) = mg \approx 0.98\,\mathrm{N}$, consistent with the load cell reading. Minor discrepancies between the load cell and accelerometer data could be the result to sensor noise, or differences in rotation and lateral movement of the mass over time. This sample test however was best evident of little swinging that could influence the recorded magnitude of acceleration in the Z-axis.

5 Damping Ratio Estimation via Logarithmic Decrement

5.1 Method

The damping ratio ζ was estimated from the free oscillation force data using the logarithmic decrement method, which is identical to the equation applied in Lab 5 for the cantilever beam. The zero-mean force signal $F_{\mathrm{zm}}(t) = F(t) - \bar{F}$ was formed using the steady-state mean as the equilibrium reference. Positive peaks were identified using `scipy.signal.find_peaks`. The log decrement β is the slope of $\ln(A_0/A_n)$ versus cycle number n. Pulling the derivations of the log decrement equation, the following equations for the amplitude ratio and damping ratio are shown below:

$$\ln\left(\frac{A_0}{A_n}\right) = \frac{2\pi\zeta}{\sqrt{1-\zeta^2}} \cdot n = \beta \cdot n \tag{8}$$

$$\zeta = \frac{\beta}{\sqrt{4\pi^2 + \beta^2}} \tag{9}$$

5.2 Results

Figure 5 shows the recorded data with detected peaks. These detected peaks are then transferred for calculation of Figure 6, which finds the logarithmic decrement data and linear regression. The damped period was estimated from the mean peak spacing over all 67 detected positive peaks.

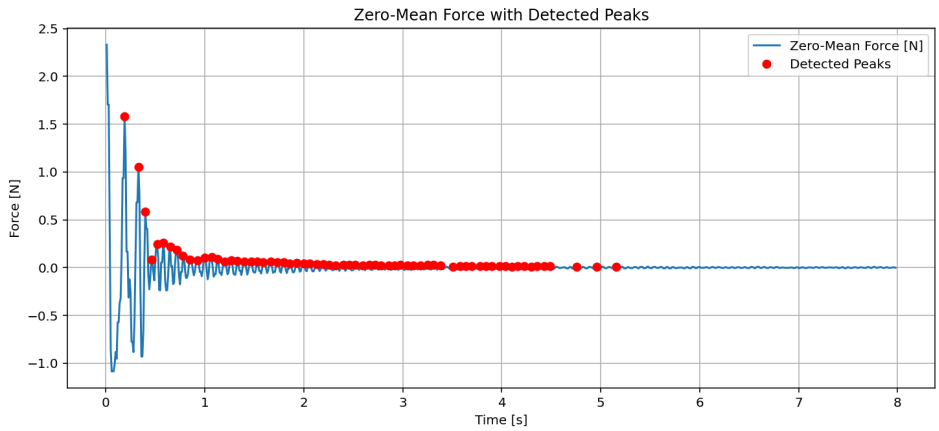

Figure 5: Force vs. Time with Detected Peaks

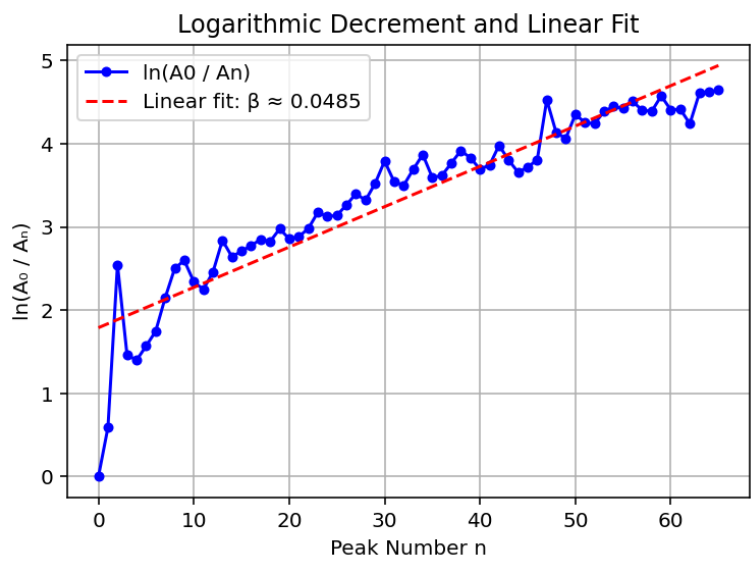

Figure 6: Log Decrement $\ln(A_0/A_n)$ vs. Cycles with Linear Regression. $\beta = 0.0485$, $R^2 = 0.858$.

The resulting dynamic parameters are summarized in the table below

Table 5: Summary of Estimated Mass-Spring-Damper System Parameters from Log Decrement Analysis

Parameter	**Symbol**	**Value**
Log decrement (slope)	β	0.0485
Regression R^2	R^2	0.858
Damping ratio	ζ	7.714×10^{-3}
Damped period	T_d	0.0743 s
Damped natural frequency	ω_d	84.5561 rad/s (13.4575 Hz)
Undamped natural frequency	ω_n	84.5586 rad/s (13.4579 Hz)
Effective spring stiffness ($k = \omega_n^2 m$)	k	714.6 N/m
Object mass	m_{mass}	99.94×10^{-3} kg
Bungee mass	m_{bungee}	10.81×10^{-3} kg

5.3 Discussion

The estimated damping ratio $\zeta \approx 7.7 \times 10^{-3}$ is very small, consistent with the lightly damped oscillation observed in Figure 3. At low damping, $\omega_d \approx \omega_n$, which is shown by the equation for ω_d as function of damping ratio and natural frequency:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \approx \omega_n, \quad \zeta \ll 1 \tag{10}$$

The effective stiffness back-calculated from $k = \omega_n^2 m = (84.6)^2 \times 0.09994 = 714.6\,\mathrm{N/m}$ represents the stiffness of the bungee cord at the operating point near the 100 g equilibrium. This value is considerably lower than the initial large-displacement slope computed from the static data in Section 2. This should be somewhat expected because the bungee cord is significantly stiffer at small stretches (refer to the steep increase from 0 to 10 mm strain in Figure 2) than in the elastic plateau region where the dynamic oscillations take place. The moderate $R^2 = 0.858$ on the log-decrement regression suggests that the damping may not be perfectly linear, with the early high-amplitude cycles decaying quickly comparatively to later cycles.

For data validation purposes, the extracted slope for small deformations from 0 to 20 mm of the bungee cord from Figure 2 indicate a linear regression slope, or k value, of 182.28 N/m, a near 5× increase in stiffness of the full plot. This further shows that the bungee elastic behavior is extremely non-linear, and given the small oscillation test was run at a starting amplitude of $\approx 1/2$ inch (12.7 mm), this non-uniform stiffness is closer to the back-calculated stiffness of the experiment.